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Entrepreneurial Learning and Critical Reflection

Discontinuous Events as Triggers for 'Higher-level' Learning

Abstract *The importance of 'learning events' has become an emergent theme within theorizing on how entrepreneurs learn. This article builds a deeper understanding of the learning outcomes triggered by significant, discontinuous events during the entrepreneurial process. It suggests that the domain of entrepreneurship represents a special and unique context in which to study management learning. It is argued that there is more to learning from discontinuous events than the incremental accumulation of more routinized, habitual, 'lower-level' learning. This article illustrates that these events have the capacity to stimulate distinctive forms of 'higher-level' learning—learning that is fundamental to the entrepreneur in both personal and business terms. It goes on to explore the concept of critical reflection and suggests that these learning outcomes are the result of what can be described most precisely as 'inward' critical self-reflection. **Key Words:** critical self-reflection; discontinuous events; entrepreneurial learning; higher-level learning*

The Nature and Process of Entrepreneurial Learning

The aim of this article is to develop a deeper understanding of the learning experienced by entrepreneurs during the management of a small enterprise. Such a focus represents part of a wider, emerging study of the learning outcomes of entrepreneurial activity within numerous contexts, such as corporate entrepreneurship, for example (see Gibb, 2000). Concurrently, a body of theory is emerging regarding organizational learning in smaller firms (Anderson and Skinner, 1999; Penn et al., 1998; Sadler-Smith et al., 1999).

The study of how entrepreneurs negotiate the successful growth of a small business represents an unusual and special domain for the study of management learning. Of particular interest is the recognition that during the early stages of business growth the entrepreneur and the business are inextricably linked and 'the owner is still synonymous with the business' (Churchill and Lewis, 1983: 34). Over

time, however, it is apparent that many entrepreneurs become actively engaged in learning to be effective managers of people and resources. In broader terms, this represents an important aspect of the complex, intimate and dynamic relationship that exists between entrepreneurs and their businesses. It is the significant 'investment of the self', particularly the extreme levels of emotional and financial commitment and 'personal exposure' associated with small business ownership, which makes the study of entrepreneurs such a unique context within which to explore the phenomenon of learning.

In emphasizing entrepreneurial activity as a unique context for the study of management learning, the terms 'investment of the self' and 'personal exposure' require further elaboration. Many entrepreneurs commit significant personal resources to creating and managing a small venture and in so doing expose both themselves and their families to the considerable financial, emotional and social risks associated with small business ownership. As the research will demonstrate, the performance of the business is inextricably linked to the emotional and financial wellbeing of the entrepreneur, his or her family and other stakeholders. In this regard, Scott and Bruce (1987) emphasize the importance of significant events when they state that 'crises are extremely important to the entrepreneur in terms of both his business and personal life' (p. 47). Although, to some extent, the same could be said for managers of larger organizations, it is the direct, immediate and acute impact and the potential long-term consequences of discontinuous events that make entrepreneurship such an interesting area for research.

There is a widely endorsed view within the literature on entrepreneurial learning that entrepreneurs are action oriented and much of their learning, therefore, is experientially based (Rae and Carswell, 2000). It is postulated that entrepreneurs learn primarily through 'learning by doing'—encompassing such activity as trial and error, explicit problem solving and discovery (Deakins and Freel, 1998; Young and Sexton, 1997). These processes of learning, however, are not confined necessarily to entrepreneurs; rather, this notable dimension of entrepreneurial learning reflects the importance of experience as a central consideration of all learning (Boud et al., 1993). As Gibb (1997) has found,

The predominant contextual learning mode in this [small business] environment is that of . . . learning from peers; learning by doing; learning from feedback from customers and suppliers; learning by copying; learning by experiment; learning by problem solving and opportunity taking; and learning from making mistakes. (p. 19)

Accepting that entrepreneurs learn through experience, Reuber and Fischer (1993) argue that it is the need to understand more deeply *how* entrepreneurs learn from formative experiences, through the identification of entrepreneurial learning 'mechanisms', which remains a key research objective. More recently, several theorists have risen to this challenge and there is growing recognition that certain 'episodes' or 'events' have an important role to play in the process of entrepreneurial learning. Rae and Carswell (2000) postulate the existence of salient 'learning episodes', where entrepreneurs have described seminal periods of learning that have been instrumental in forming their approaches to life and work. In terms of more discrete events, Young and Sexton (1997) argue that the learning activities of entrepreneurs emerge from the reactive or proactive response to opportunities and problems.

The notion of 'learning events' has become an emergent theme within theorizing on entrepreneurial learning (see, for example, Cope and Watts, 2000; Sullivan, 2000). As Deakins and Freel (1998) assert,

entrepreneurship and the growth process is essentially non-linear and discontinuous. It is a process that is characterised by significant and critical learning events. The ability of entrepreneurs to maximise knowledge as a result of experiencing these learning events will determine how successful their firm eventually becomes. (p. 153)

It is important to emphasize that entrepreneurs do not learn solely through facing significant learning 'events'; rather, they are constantly learning and developing as they manage their business (Hines and Thorpe, 1995), and can learn through routinized and habitual activity as well as more non-linear, discrete events (Costello, 1996). This more gradual, incremental aspect of entrepreneurial learning aside, there is a pressing need to develop a deeper understanding of the process of learning and the specific forms of learning associated with the management and resolution of more discontinuous events. It is with regard to these particular research issues that this article seeks to inform current debate.

The Learning Outcomes of Discontinuous Learning 'Events'

As the literature on entrepreneurial learning is still emerging, it is useful to draw on other bodies of literature to inform this discussion of discontinuous learning events. Within the individual learning literature, although it is acknowledged that learning is a continuous process (Mumford, 1991), it seems that challenges and problems can be rich sources of learning (Daudelin, 1996; Kleiner and Roth, 1997). Furthermore, it is apparent that learning from more discrete and unusual events can often be 'transformational' (Appelbaum and Goransson, 1997), in the sense that when individuals face such non-routine situations their learned responses and habitual ways of behaving prove ineffectual (Marsick and Watkins, 1990).

Such exceptional circumstances require heightened attention and experimentation, forcing individuals to question their taken-for-granted beliefs and assumptions and reframe their understanding of the situation at hand (Argyris and Schön, 1978; Schön, 1983). From an organizational learning perspective, Nicolini and Mesnar (1995) refer to such events as 'discontinuities' or 'breakdowns' in the flow of organizational functioning:

The identity of the organization after the breakdown is different from before. The difference, and the expression of good intention regarding the effort to not incur in the same breakdown again, are called 'what we have learned'. (p. 739)

One of the most intriguing aspects of discontinuous learning events, such as opportunities or crises, is the notion that these incidents can stimulate different 'levels' of learning. Fiol and Lyles (1985) emphasize that some kind of 'crisis' is a prerequisite for a challenging and fundamental form of learning, which they describe as 'higher-level' learning. As Fiol and Lyles (1985) go on to state, such 'shocks', 'jolts' or 'crises' are necessary 'for unlearning, new higher-level learning and readaptation to take place' (p. 808). Similarly, it is argued later in this article that

there is more to entrepreneurial learning from discontinuous events than the gradual, incremental accumulation of lower-level, 'instrumental' learning (Mezirow, 1990).¹ It is suggested that entrepreneurs can experience *distinctive* forms of 'higher-level' learning from facing, overcoming and reflecting on significant opportunities and problems during the entrepreneurial process. Before embarking on an explanation of this assertion, it is first necessary to clarify what is meant by the notion of different 'levels' of learning.

'Higher-level' versus 'Lower-level' Learning

The development of a hierarchy of learning levels is an important outcome of inquiry into both individual and organizational learning. Interestingly, many of the concepts developed to help describe the phenomenon of learning, such as the notions of different 'levels' of learning (Fiol and Lyles, 1985) and 'mental models' (Senge, 1990), could be described as anthropomorphic. This is because terms developed to describe how individuals learn are also used to describe how organizations are thought to learn, and vice versa. As Argyris and Schön (1978) recognize, 'it is true that we do apply to organizations many of the terms we also apply to individuals' (p. 11). It must therefore be clarified at the outset that certain theoretical terms used in the study of organizational learning will be employed to describe the individual learning outcomes of discontinuous events experienced by entrepreneurs.

Different levels of learning distinguish between more practical, routine, adaptive learning and more fundamental learning that generates new understandings and new cognitive 'theories for action' (Argyris and Schön, 1978)—thereby forcing individuals to question their established ways of doing things (Huber, 1991). Kim (1993) argues that there are two facets or meanings to the term learning: 'operational' and 'conceptual'. Operational learning regards the acquisition of 'know-how', i.e. what people learn, as opposed to the acquisition of 'know-why', which refers to how individuals develop a conceptual understanding of an experience and subsequently apply this learning.

The literature on individual and organizational learning is replete with such dualistic conceptualizations of how people are thought to learn. This dichotomy has been encapsulated within such terms as 'lower-level' and 'higher-level' learning (Fiol and Lyles, 1985); 'surface' and 'deep' learning (Brown, 2000); 'adaptive' and 'generative' learning (Gibb, 1995; Senge, 1990); 'incremental' and 'transformational' learning (Appelbaum and Goransson, 1997) and 'instrumental' and 'transformative' learning (Mezirow, 1990). Probably the most popularized notion of two learning 'levels' is the work of Argyris and Schön (1974, 1978), who distinguish between 'single-loop' and 'double-loop' learning. Single-loop learning is depicted as a more adaptive form of learning, which may challenge and thereby 'correct' or 'modify' an individual's (or an organization's) existing strategies and assumptions. However, the underlying 'theories for action' that influence and shape these strategies remain unchanged. In contrast, double-loop learning has the capacity to fundamentally change an individual's theories for action, thereby creating entirely new assumptions and strategies for effective action.²

Although the terminology used to describe these dual elements of learning can vary within the learning literature, the primary features that distinguish between these two different levels of learning are often very similar. For certain theorists, the various descriptions mentioned are even viewed as synonymous, with the terms 'higher-level', 'generative', 'transformational' and 'double-loop' being used interchangeably (Appelbaum and Goransson, 1997; Nevis et al., 1995). Fiol and Lyles (1985) provide a definition of both 'lower-level' and 'higher-level' learning. These definitions encapsulate many of the distinctions raised by other learning theorists and represent useful working definitions in this article when discussing the learning outcomes of discontinuous events.

Lower-level learning: Focused learning that may be mere repetition of past behaviours—usually short-term, surface, temporary, but with associations being formed. Captures only a certain element . . . Single loop, routine level.

Higher-level learning: The development of complex rules and associations regarding new actions. Development of an understanding of causation . . . Double loop learning. Central norms, frames of reference and assumptions changed. (Fiol and Lyles, 1985: 810)

It is important to highlight that the notion of 'higher-level' or 'higher-order' learning is used here as a generic descriptor to encompass numerous forms of learning described in different bodies of literature. Thus the concepts of 'double-loop', 'transformative' and 'generative' learning, for example, are described here as distinctive forms of higher-order learning. In summary, Table 1 outlines some of the primary issues that distinguish lower-level learning from more fundamental, higher-level learning.

Of particular significance is the recognition that higher-order forms of learning have the capacity to challenge or redefine what has been described variously as an individual's 'mental models' (Appelbaum and Goransson, 1997; Kim, 1993); 'frames of reference' (Huber, 1991) or 'theories for action' (Argyris and Schön, 1978). As Senge (1990) explains,

'Mental models' are deeply ingrained assumptions, generalisations, or even pictures or images that influence how we understand the world and how we take action. Very often, we are not consciously aware of our mental models or the effects they have on our behaviour. (p. 8)

In essence, a mental model represents a representation or simplification of an individual's view of the world, which includes their knowledge, beliefs and experiences as well as their implicit and explicit understandings (Kim, 1993; Spicer, 1998). The notion of 'mental models' also bears a striking resemblance to the concept of 'meaning perspectives' put forward by Mezirow (1991). As Mezirow (1991) explains, an individual's meaning perspectives represent perceptual and conceptual frameworks that 'form, limit, and distort how we think, believe, and feel and how, what, when and why we learn' (p. 34). Thus it is the recognition of cross-situational significance, particularly the notions of transferability and applicability, which set higher-level forms of learning apart, i.e. the ability of such learning to shape the behaviour and actions of individuals when they are confronted with new experiences, situations and contexts.

Table 1 Distinguishing features of lower- and higher-level learning

Contributing Theorist(s)	Lower-level learning	Higher-level learning
Gibb (1995)	Learning in order to cope with change and survive	Learning that involves the capacity to 'bring forward' experience
Huber (1991)	Learning within a 'frame of reference'	Learning a new 'frame of reference'
Argyris and Schön (1978)	'Single-loop' learning regards routine, immediate tasks	'Double-loop' learning regards the questioning of underlying values which guide action; implies an awareness of long-range outcomes
Senge (1990)	'Adaptive' learning involves coping with the current environment in new and better ways (cited in Sadler-Smith et al., 1999) ^a	'Generative' learning moves beyond adaptation, requiring individuals and organizations to develop new ways of looking at the world (cited Sadler-Smith et al., 1999)
Mezirow (1990, 1991)	'Instrumental' learning is involved in task-oriented problem solving—how to do something or how to perform. Regards developing an understanding of the procedural assumptions guiding the problem-solving process	'Transformative' learning has the capacity to transform an individual's 'meaning perspectives'—perceptual and conceptual frameworks that form, limit and distort how individuals think, believe, feel and what, when and why they learn
Pask (1976)	' <i>Serialist</i> ' strategy involves detailed, step-by-step approach from one idea to the next without necessarily considering the whole picture	' <i>Wholist</i> ' strategy involves learning in relation to the whole
Fiol and Lyles (1985)	Occurs through repetition and routine, short-term outcomes	Has long-term effects; more of a cognitive process that involves skill development and new insights
Appelbaum and Goransson (1997)	'Adaptive' learning involves more mundane, everyday, incremental learning	'Transformational' learning involves radical change; learning that requires a shift in 'mindset'

Note: ^a for the purposes of this table, Sadler-Smith et al. (1999) are cited here as they provide useful and succinct definitions of adaptive and generative learning that are not clearly apparent in Senge's (1990) original work.

In conceptualizing discontinuous events as a significant entrepreneurial learning 'mechanism', it is useful to draw on theories of higher- and lower-level learning to inform current understanding of these events within the fields of entrepreneur-

ship and management. More specifically, the capacity for these events to trigger 'higher-level' learning is of particular importance, given the complex relationship between such learning and subsequent action. It was within this theoretical context that the research informing this discussion was conducted.

It is necessary to highlight at this point that defining discontinuous events as higher-order learning events is a subjective and interpretive value judgement. Although learning levels are often presented in discrete, dichotomous terms, it is important to remember that they are actually parts of a continuum (Argyris and Schön, 1978). Sadler Smith et al. (1999) point out that differentiating between different levels of learning is often very difficult, as they remain inextricably linked and indistinct. As a result, 'identifying where adaptive learning stops and generative learning starts is difficult and often relies to a certain extent upon the subjective assessment of the analyst' (Sadler-Smith et al., 1999: 881). Consequently, it is seen as vitally important that some level of transparency is maintained in this article in terms of illustrating how the author has arrived at the findings presented here and that the reader is able to appreciate the context in which these events have been conceptualized as higher-level learning experiences. For this purpose, detailed case examples will be introduced, an issue discussed in more detail shortly.

The Research

In understanding the learning outcomes of discontinuous events, this article is based on qualitative, case study research into the 'lived experience' (Thompson et al., 1989) of six practising entrepreneurs. The focus of the research was the learning and personal development experienced by entrepreneurs within the context of small business ownership and growth (Cope, 2001). Within the overall research strategy of the case study (Stake, 1995), in-depth, unstructured 'phenomenological interviews' (Thompson et al., 1989) were conducted with the six participants, which explored the developmental history of their businesses.³ The research questions shaping this inquiry were as follows.

1. How do entrepreneurs perceive, think about and give meaning to the process of their personal learning over time?
2. How do entrepreneurs feel that they have changed since they entered into small business ownership?
3. What role do critical incidents have to play within entrepreneurial learning and development?
4. What are the potential dynamics of the relationship between organizational growth and entrepreneurial learning and development?

The exploration of critical incidents was an integral part of the interview process, primarily because these discontinuous events were perceived as a 'way in' to studying the complex and dynamic phenomenon of entrepreneurial learning. The interviews focused on these 'memorable' moments (Chell, 1998) from a number of different perspectives—the antecedents of the event; how it was managed and resolved; the impact and significance of the event; and, with

hindsight, how it may have been handled differently. It was envisaged that this reflective process would enable the learning outcomes of the events to be explored.

Although the approach used in the research had similarities to the 'critical incident technique' (Flanagan, 1954), a method utilized quite recently within the field of entrepreneurship (Chell, 1998), this technique was seen as too structured for the purposes of the research. Rather than asking the participants to select three critical incidents, two positive and one negative, at the beginning of the interview, as Chell (1998) recommends, critical events were discussed more informally during the interview. The participants were simply asked to focus on the best and worst times that they had experienced since they had started their businesses as they described how their businesses had developed. It must be emphasized that the focus of the research was *self-defined criticality*, as it was the participant's personal representation of salient moments that was of prime importance.

In exploring these events, the role of the researcher during the interview process was not, however, a passive one. Although the identification and initial narration of significant incidents were emergent and participant-driven, naturally taking the form of stories (Gilbert and Morris, 1995; Gold et al., 2002), the researcher then played an active role in helping the participants to reflect more deeply on these occurrences by focusing on the perspectives outlined earlier. In this respect, the interview itself was, to some extent, a sense-making process in which, together, the researcher and the participant clarified the meanings and importance attached to their experiences.

Using a very different approach, the research reported by Gold and Holman (2001) and Gold et al. (2002) develops some interesting and valuable insights into stimulating such critical thinking and reflection with regard to significant incidents. During a personal development module for managers, the authors encouraged participants to use 'story logs', where stories of salient events were collected. Using Toulmin's (1958) approach to argumentation, the authors were able to help managers explore prominent work experiences and facilitate 'new understandings and insights (especially with regard to their identity), new perspectives and new ways of acting' (p. 372).

From the rich and diverse array of discontinuous events narrated by the participants, two case examples will be introduced.⁴ The events described in these case examples are deemed to be extremely important in theoretical terms for two primary reasons. First, they illustrate that non-routine events have a prominent role to play in how entrepreneurs learn. Second, they demonstrate two *distinctive* forms of higher-level learning that can be triggered by discontinuous events during the management of a small enterprise. Thus the learning outcomes of these events can be understood in relation to two conceptual frameworks that describe different forms of higher-order learning—the work of Argyris and Schön (1978) on 'double-loop' learning and that of Mezirow (1990, 1991) on 'transformative' learning.

The decision to focus on these two theories of learning, in particular, was based on two key factors. First, it was perceived that these concepts provide well-articulated and informative theoretical frameworks that demonstrate strong explanatory power in relation to this research—particularly in terms of describing

the forms of learning that occurred in relation to the two discontinuous events introduced here. Second, these two theories illustrate different yet complementary facets of higher-level learning, which together enable both a deep and a broad understanding of the kinds of learning that can be triggered by discontinuous events.

Although a more in-depth discussion of double-loop and transformative learning will emerge in the discussion of the two case examples, it is useful to clarify briefly why these two forms of learning are both of a higher order and yet remain qualitatively different. Interpreting these conceptual frameworks in relation to this research, from an individual perspective it appears that double-loop learning involves the entrepreneur developing a renewed understanding or redefinition of the *organizational* processes and strategies that are employed within his or her organization. In contrast, transformative learning has a distinctly personal dimension and has the capacity to stimulate considerable changes in the entrepreneur's *self-understanding*. As Mezirow (1991) states, transformative learning involves 'an enhanced level of awareness of the context of one's beliefs and feelings [and] . . . involves profound changes in self, changes in cognitive, emotional, somatic, and unconscious dimensions' (pp. 161, 177). It is these clear distinctions between double-loop and transformative learning that will be demonstrated in the first case example, which is introduced next.

Case Example 1: Andrew

Andrew is the owner of a company that specializes in the provision of language communication skills training for managers and professionals in Europe, particularly Scandinavia. At the time of the events discussed in this case example, the company had been in existence for approximately six years and Andrew employed four full-time trainers. This case example includes a number of non-routine events that occurred within a wider episode of upheaval and internal conflict between different members of the organization.⁵

This period of internal disharmony occurred as the business was starting to enjoy rapid growth. Andrew was still occupying a number of different roles and trying to manage all the activities within his business. As such, the business still had very informal operating procedures and job roles. As Andrew recalls, 'all these people were running around and they hadn't got a bloody clue what was going on and so they had to turn to me for everything' (Andrew, transcript 1: 11). Eventually, this situation led to a heated confrontation with one member of staff, who openly challenged the informal responsibilities that Andrew expected his staff to fulfil. It was after this unexpected discontinuous event that Andrew recognized he was in what he describes as a 'sticky situation'. In retrospect, Andrew acknowledges that it was this initial altercation that sparked the prolonged period of internal conflict within the organization.

The disagreement that Andrew experienced with one of his employees also occurred at an important time in the endogenous development of the business. Not only did Andrew's father retire, a person whom Andrew described as a 'very stabilizing force' within the organization, but around the same time Andrew recruited a general manager. It was within this wider phase of internal change and

disruption that the conflict between Andrew and his employees intensified. The dispute spread to the rest of his employees, many of whom took opposing sides, and in-house quarrelling became more widespread. Consequently, this internal disharmony became unbearable for Andrew and it was during a holiday in Ireland that he thought intensely about how to remedy this increasingly unproductive situation. As he explains,

I sat for about two or three hours every afternoon thinking *what do we do, how do we go on?* And I came to the conclusion that we had to put systems in place, although I didn't know how to . . . Business Link came to do a sort of SWOT . . . analysis and they pointed out the fact that there weren't any systems in place, people didn't even have a pay slip. I mean looking back on it now, half of what I was doing was illegal, but it wasn't devious, just lack of knowledge. (Andrew, transcript 1: 16, emphasis added)

Although the conflict that Andrew experienced with his employees proved very painful and difficult at the time, it had very positive longer-term consequences for both Andrew and his organization. Ultimately, it marked the beginning of a wider transition towards a more professionally managed enterprise, where clearly identifiable structures, systems and responsibilities were gradually put in place. As Andrew states 'that conflict was the start of the properly organized, administrated, well run business we've got today' (Andrew, transcript 1: 23).

On reflection, it is apparent that Andrew has learned some important lessons from this episode and has experienced a significant change in self-awareness. With hindsight, Andrew recognizes that he would now do things very differently.

Well you can either change people or you move them on. What I would have done is . . . I'd have been hard. I'm a great optimist and [I was] naively optimistic and so I thought maybe we can make it work . . . Wrong . . . So now, that's wisdom, you talk about wisdom . . . but what I've learnt is deal with it immediately, then that way people can be quite clear about how the land lies. (Andrew, transcript 1: 30)

Case Discussion

Relating these events to the notion of double-loop learning, to use Argyris and Schön's (1978) terminology, it seems that for the first time an employee openly questioned a tacit and previously accepted 'norm' of how staff were meant to behave within Andrew's business; a norm that was part of the implicit *theory-in-use* inculcated by Andrew within the organization.⁶ Andrew's actions do not reflect the simple implementation of a 'corrective' measure to maintain some kind of stability and consistency within the business and 'keep things as they are'—an action that is indicative of 'single-loop' learning (Argyris and Schön, 1978). As Argyris and Schön (1978) state, single-loop learning involves 'learning episodes which function to preserve a kind of *constancy*' (p. 18, emphasis added). Sacking the employee and carrying on in the same way without questioning the underlying reasons why this event occurred would have been a more obvious single-loop learning outcome.

Instead, Andrew acknowledged that a major conflict had been triggered within his business that could not easily be resolved. As Argyris and Schön (1978) assert, incompatible requirements in an organization's theory-in-use are characteristically

expressed through a conflict between members of the organization; a situation demonstrated quite clearly in this case example. It is the recognition of such conflict that is the first step in a manager's 'double-loop' learning process:

[Managers] must reflect on this error to the point where they become aware that they cannot correct it by *doing better what they already know how to do* . . . Then the managers must undertake an inquiry which resolves the conflicting requirements. The results will take the form of a restructuring of organizational norms, and very likely a restructuring of strategies and assumptions associated with those norms, which must then be embedded in the images and maps which encode organizational theory-in-use. We call this sort of learning *double-loop*. (Argyris and Schön, 1978: 22)

The learning that Andrew experienced from reflecting on the confrontation with one of his employees and the resultant period of internal conflict could be described as double-loop because it involved considerable organizational *change*. Andrew seriously questioned the state that his business was in—a period of reflection that provided a potential solution to the disharmony and eventually led to the comprehensive restructuring of the organization. As Andrew openly admits, he was naive with regard to putting systems and structures in place and this represented a process that *he did not know how to do*.

Although Andrew experienced higher-level, specifically 'double-loop' learning about his organization, it is also apparent that Andrew experienced fundamental personal learning about himself as a result of the dispute with his employee. In particular, he realized that he was naively optimistic and 'too soft' as a small business owner—aspects of his persona that he has now learned were detrimental to the success of his business.

This case example raises some important issues regarding management learning in smaller firms. The problems that Andrew experienced with his employees are a clear indication of the common difficulties that small business owners often have in learning to become effective managers of people and resources (Burns and Harrison, 1989; Hofer and Charan, 1984). As Andrew states, 'I think it was this painful process of learning how to delegate. You delegate and then you immediately look for ways in which the person has carried out the task differently and not as well as you might have done' (Andrew, transcript 1: 19). For Andrew, it was not only a process of learning to become an effective manager but also recognizing his limitations as a manager. Consequently, Andrew has renegotiated his own role within the organization and adopted a less hands-on approach to managing his employees. In describing his relationship with his general manager, Andrew states that 'I've learnt the difference between leadership and management now basically. I'm a leader and he is the manager. I can't manage and he can't lead' (Andrew, transcript 1: 47).

Interpretation suggests that this change in self-awareness that Andrew experienced was what Mezirow (1990, 1991) describes as 'transformative', as this realization of his own abilities and limitations allowed him to find some kind of personal direction and enabled him to relax and enjoy being a small business owner again. As he states,

I feel better. I mean and before I used to have dreadful problems with muscles in the back of my neck and dreadful problems with acid, you know, on the way to an ulcer. I

don't get anything like that at all any more. I just feel better in myself, my lifestyle is considerably better . . . so I don't worry anymore. (Andrew, transcript 1: 12)

In conceptualizing the conflict that Andrew experienced with his employees as both a double-loop and a transformative learning experience, it is important to emphasize that although these forms of learning represent *distinctive* examples of 'higher-level' learning, they are by no means mutually exclusive. As this case example illustrates, with regard to learning from discontinuous events they can be overlapping concepts. As a result, not only is deciding between different levels of learning a complex process (Sadler Smith et al., 1999), it is suggested here that differentiating between different *forms* of higher-level learning can also be problematic. As the individuals discussed here are entrepreneurs who are learning within the emotive context of small business ownership, it is often hard to differentiate clearly between 'double-loop' and 'transformative' learning outcomes. Accepting that the quality of higher-level learning may vary (Argyris and Schön, 1978), it is suggested that the 'richness' of the learning experience may well be greater if discontinuous learning events trigger both double-loop and transformative learning outcomes—learning that is fundamental in both organizational and personal terms.

The Role of Discontinuous Events in 'Transformative' Personal Learning

In understanding the deeply personal learning that entrepreneurs may encounter as they manage their businesses, it is useful to explore further the process of adult learning described by Mezirow (1990, 1991) as 'transformative' learning. As Mezirow (1991) explains, transformative learning 'involves a sequence of learning activities that begins with a disorienting dilemma and concludes with a changed self-concept that enables a reintegration into one's life context' (Mezirow, 1991: 193). What is so interesting about this concept of transformative learning is the explicit recognition that this higher-order learning process is triggered by some kind of discontinuous event, described by Mezirow (1991) as a 'disorienting dilemma' or 'crisis'. As Mezirow (1990) states,

perspective transformation occurs in response to an externally imposed disorienting dilemma—a divorce, death of a loved one, change in job status, or other . . . Anomalies and dilemmas of which old ways of knowing cannot make sense become catalysts or 'trigger events' that precipitate critical reflection and transformations. (Mezirow, 1990: 14)

These 'disorienting dilemmas' described by Mezirow (1991) are primarily events that happen in the personal life of the individual concerned. What makes the study of entrepreneurs such an unusual and important context for the study of management learning is that the majority of these individuals own and manage their own businesses. Consequently, the emotional complexity and intimacy of the relationship between the entrepreneur and the small business is unique. As another of the six research participants, Luke, emphasized, 'I look at myself and the business as one, because the business is me and I am the business'.⁷ The

inextricable link between the individual and the business adds an interesting insight into the process of transformative learning. This is because the experiences of the six participants studied in the original research (Cope, 2001) illustrate that *organizational* discontinuous events have an important role to play in stimulating transformative personal learning, as will be illustrated in the second case example introduced shortly.

The present research also suggests that within the context of entrepreneurship, such 'trigger events' are not always 'externally imposed' as Mezirow (1990) suggests. Rather, the catalyst for transformative learning can sometimes be an event that is largely *self-imposed* and is the result of a clear mistake made by the entrepreneur. This is particularly pertinent with respect to more serious mistakes that trigger an organizational crisis, where the actions of the entrepreneur stimulate a discontinuous event that impacts at all levels of the organization, as is the situation in the second case example. In this respect, the research reported here reinforces Mezirow's (1990, 1991) assertion that crises are powerful stimulants of transformative learning. However, interpretation suggests that it was those events that the research participants felt were self-generated and were *not* beyond their personal control or influence that tended to be more salient and transformative personal learning events. These research findings regarding perceived personal responsibility and associated learning have wider significance for the study of management learning. An interesting area for further research would be to consider the extent to which this finding is unique to entrepreneurs, or may also be applicable to managers of larger organizations.

Andrew's case example introduced the notion that both double-loop and transformative learning outcomes can be triggered by a discontinuous event. The second case example which follows, Simon's, is included here because of all the events described by the six participants in the research (Cope, 2001), this is the most convincing and illustrative example of the kind of 'transformative' learning that Mezirow (1990, 1991) describes. If Simon's experiences are interpreted from an adult learning perspective, it is apparent that the learning that Simon engaged in was rather special, unusual and 'holistic' (Boud et al., 1985b). Drawing on the work of Boud et al. (1985a, b), theorists who have written extensively on the nature and process of adult learning, the following comment best describes the kind of learning that Simon experienced:

Some learning can become so related to the self that it enters into our sense of identity and can have considerable importance and become a significant force in our lives. Significant feelings can come to be attached to this type of learning and any learning experience which touches this area can give rise to strong emotions. (Boud et al., 1985b: 33)

Case Example 2: Simon

Simon has had a complex managerial and entrepreneurial career. After building a career in corporate finance, Simon had started and acquired numerous small ventures both prior to, and following, the discontinuous event described in this case example. The event described here occurred within a radio communications

company that Simon had taken control of less than a year before, after forming an equal partnership with the existing owner-manager of the business.

The dilemma that Simon faced was undoubtedly one of the most serious crises that can face any small business owner—potential bankruptcy. This crisis was self-imposed, as Simon acknowledges that he made a serious and very basic marketing error that he thought he was not capable of making, a mistake which effectively ruined the company. With hindsight, Simon does not know how he would handle such a stressful period now, stating that ‘it was the end. If I got within two miles of that now . . . I don’t know how I’d cope with it, but then I was different’ (Simon, transcript 1: 28).

Simon made the initial marketing mistake just before the Christmas holidays. As a result, he was able to think about what to do during this brief interlude and take preventative action. Simon managed to engineer a solution that allowed him to walk away from the situation, while leaving his partner with a failing company that was clearly heading for bankruptcy. Simon describes this discontinuous event as a ‘moral dilemma’, as everything was on the line and he had to put his wife and family first. As Simon stated, there was no easy way out of the situation, he had the ability to save himself and he took it.

It is apparent that the way in which Simon saved himself during this traumatic event represented both the essence and ultimate test of his entrepreneurial skills. Of particular significance is the fact that Simon openly recognizes that prior to this particular crisis, facing such risky situations was what he was good at and enjoyed doing. As he stated, ‘I like the challenge of facing the possibility of defeat. I think that’s probably one of the reasons why I like duff companies . . . when it’s failing and the previous people have made a muck of it, can I do something with it?’ (Simon, transcript 1: 26). What was so different about this event, however, was that for the first time Simon learned that he was not infallible.

I mean that was a seminal, you asked for seminal moments, that was the first time in my business career that I really realized how easy it was for me to make a mistake and since then I have been considerably more cautious . . . I realized at that point that I wasn’t immune to disaster. (Simon, transcript 1: 29)

One reason why this discontinuous event had such a dramatic personal effect on Simon was that the consequences of his mistake crossed the important boundary from the business realm to the personal, as the event endangered his wife and children as well as himself. A further reason why this event proved to be so painful is that Simon realized exactly how single-minded and ruthless he could be in a time of crisis, which he describes as a ‘frightening’ realization that he has had to live with ever since. ‘I found that quite a difficult experience, but I realized there were certain things that I just wouldn’t allow to come up’ (Simon, transcript 1: 29). What is also apparent is that this event proved to be a deeply personal learning experience for Simon, radically changing both his attitude to entrepreneurship and his perceptions of himself.

I’m much more acutely aware of trying to look after everyone and now I’m far more inclusive because it’s almost as if I felt that was a sort of judgement day . . . it was, arguably . . . a turning point and from that point onwards I’ve looked very differently at things and tried to avoid such major moral positions like that because there was no right

and wrong, something had to go somewhere and I chose my wife and children. (Simon, transcript 1: 31)

Case Discussion

It is clear from these reflections that this discontinuous event had important personal consequences for Simon. The last comment, in particular, illustrates that this event permanently changed him in some way and represented an important turning point in his career as an entrepreneur. Examined from a theoretical perspective, it seems that Simon engaged in higher-level, 'transformative' personal learning during this event (Mezirow, 1991). This is because he faced a 'moral' disorienting dilemma that brought into question his notions of right and wrong, thereby forcing him to question his 'meaning perspectives' as a business owner. As Mezirow (1991) explains, such meaning perspectives

provide us with criteria for judging or evaluating right and wrong, bad and good, beautiful and ugly, true and false, appropriate and inappropriate. They also determine our concept of personhood, our idealised self-image, and the way we feel about ourselves. (p. 44)

Simon's meaning perspectives were brought into sharp relief during this event, as they proved ineffective when trying to decide on the notions of right and wrong. Although he came to no clear conclusion during the event, it seems that he now feels very differently about himself and his role as the owner and manager of a small business, particularly in terms of the level of 'personal exposure' that he is willing to accept.

To summarize, within the entrepreneurial learning literature, the management and resolution of discontinuous events has been identified as a significant learning mechanism for entrepreneurs (Cope and Watts, 2000; Deakins and Freel, 1998; Sullivan, 2000). There is still much to discover, however, both about the process of learning and the specific forms of learning associated with these non-routine events (Cope, 2001). These two case examples begin to shed some light on these issues, demonstrating that entrepreneurs can experience distinctive forms of higher-level learning as a result of facing discontinuous events. In particular, these case examples illustrate that entrepreneurs can experience 'double-loop' learning (Argyris and Schön, 1978) about how best to manage their businesses, but also deeply personal, 'transformative' learning (Mezirow, 1990, 1991) that changes their perceptions of themselves as small business owners.

In beginning to understand how these forms of higher-level learning occur from a process perspective, it seems that there is a 'critical' dimension to double-loop and transformative learning outcomes triggered by discontinuous events. Making sense of the work of several theorists who explore the central role of 'critical reflection' within adult learning (Boud et al., 1985a; Marsick and Watkins, 1990; Mezirow, 1990, 1991), it is apparent that the higher-order learning outcomes described here are inextricably linked with, and are the outcome of, 'critical reflection':

Critical reflection is not concerned with the how or the how-to of action but with the why, the reasons for and consequences of what we do . . . By far the most significant

learning experiences in adulthood involve critical self-reflection—reassessing the way we have posed problems and reassessing our own orientation to perceiving, knowing, believing, feeling, and acting. (Mezirow, 1990: 13)

The 'Critical' Dimension of Learning from Discontinuous Events

The complex and disputed meanings and connotations attached to the term 'critical' means that due care needs to be taken when applying this concept. Willmott (1993) argues that diverse, constructed and contextual meanings have been attributed to the term 'critical thinking' and the term varies depending on the 'framework of interpretation' favoured by different commentators. Certain writers would argue, for example, that the learning and reflection described by Mezirow (1990, 1991) is not really 'critical', as it is limited to personal and psychological, rather than social transformation (Reynolds, 1998). It is important, therefore, to be clear about the type of 'critical reflection' discussed in this article. Interpretation suggests that the reflection that the participants engaged in was not 'critical' from the Critical Theory perspective, for example. This is because it was not concerned with challenging power relations in the pursuit of emancipatory social and political ideals (Cavanaugh and Prasad, 1996; Kemmis, 1985; Willmott, 1993).

Using the 'framework of interpretation' (Willmott, 1993) adopted by adult learning theorists such as Mezirow (1990, 1991), Marsick and Watkins (1990) and Boud et al. (1985a), reflection can be critical if it challenges *personal* norms, assumptions and 'taken-for-granted'. Hence, it is vital to stress that the use of the term 'critical reflection' used here works within this more individualistic framework, encapsulating personal introspection that could be described, more specifically, as 'inward' critical self-reflection (Kemmis, 1985).

It is, perhaps, the continued use of the term 'critical' by authors such as Mezirow (1990, 1991) and Boud et al. (1985a) that is problematic when describing such personal reflection. The terms 'deep reflection' or 'intense reflection', for example, could quite easily be substituted to describe the reflective activity of both Andrew and Simon. It is simply the notion that these individuals engaged in a form of reflection that was fundamental and deeply challenging and, therefore, significantly different from more routinized or 'simple' reflection (Marsick and Watkins, 1990). This discussion, however, must work within the restrictions of existing terminology in order to illustrate the relationship between critical *self*-reflection (Kemmis, 1985; Mezirow, 1990, 1991) and higher-level learning with regard to discontinuous events during the entrepreneurial process.

It is interesting to observe the distinctions that are apparent within even the concept of inward, critical self-reflection. Marsick and Watkins (1990) and Boud et al. (1985a), for example, describe critical reflection as a focused, explicit, goal-directed activity. As Boud et al. (1985a) state, 'reflection as we have described it is pursued with intent. It is not idle meanderings or daydreaming, but purposive activity directed towards a goal' (p. 11). Returning to Andrew's case example, his comments suggest that he did engage in more structured and proactive reflection. The statement that he sat 'for about two or three hours every afternoon thinking what do we do, how do we go on?' (Andrew, transcript 1: 16) is a powerful

example of a more focused analysis of his business that generated 'double-loop' learning outcomes (Argyris and Schön, 1978).

Interpretation suggests that the reflection that Andrew and Simon engaged in was also deeply personal, as they critically analysed their own perceptions and assumptions and challenged the efficacy of their actions and behaviour. This is more akin to the notion of critical reflection as described by the work of Mezirow (1990, 1991), which describes this fundamental form of reflection as an activity that challenges an individual's 'self-concept' and leads to 'transformative' personal learning. This is clearly apparent in Simon's case example, as he learned some important lessons about himself from reflecting on the traumatic experience that he faced, thereby significantly changing his perceived self-image. The notion of critical reflection put forward by Mezirow (1990, 1991), therefore, has the capacity to stimulate fundamental personal learning and a more enlightened understanding of *oneself*.

The present research suggests that the concept of more structured, goal-directed critical reflection, as proposed by Boud et al. (1985a) and Marsick and Watkins (1990), can have very different learning outcomes. In a similar fashion to Mezirow's work (1990, 1991), their depiction of critical reflection is confined, primarily, to a discussion of critical self-reflection that challenges personal assumptions. However, as Andrew's case example illustrates, such purposive critical reflection can stimulate 'double-loop' learning outcomes, which enable a deeper understanding of one's *business*.

Conclusion

Theories of entrepreneurial learning are still emerging, but there is growing recognition that the management of a small enterprise is a turbulent and non-linear process, which is characterized by salient 'learning events' (Deakins and Freel, 1998). This article has sought to develop a deeper understanding of the specific learning outcomes that can be triggered by discontinuous events during the entrepreneurial process. It has provided evidence that these non-routine events represent a key entrepreneurial learning mechanism, involving much more than the gradual accumulation of more routinized and habitual 'lower-level' learning (Fiol and Lyles, 1985). Rather, the two case examples introduced have illustrated that discontinuous events can stimulate distinctive forms of higher-order learning that prove to be fundamental in both personal and business terms.

More research is needed to explore the social dimensions of the learning process associated with discontinuous events, particularly the 'active encounters' (Burgoyne, 1995) that entrepreneurs may engage in, both during and after these events. Preliminary findings by Taylor and Thorpe (2000) suggest that network interactions during discontinuous events can stimulate reflection and learning. Similarly, Sullivan (2000) has reported some success in facilitating higher-level learning from critical incidents using previous or existing entrepreneurs as mentors. Clearly, this represents a key area for further inquiry.

In researching discontinuous events, the narration of these experiences by the participants represented a natural form of storytelling (Gilbert and Morris, 1995). An important conclusion to emerge from this research is that in order to

understand the impact and significance of these events in learning terms, a qualitative appreciation of the context in which they emerged and were resolved is crucial. In thinking about more structured approaches to stimulating higher-level learning from discontinuous events, the work of Gold et al. (2002) on critical thinking, as discussed earlier, looks extremely promising. By focusing on 'significant incidents', they report that Toulmin's (1958) framework for argumentation encouraged managers to think critically about topics and in so doing created greater levels of self-awareness among the participants. On interpretation of their work, it appears that the critical examination of topics such as current organizational practice (e.g. Why do I/we do things this way?) and the manager's own identity and life history (e.g. Why am I like this?) may be significant in stimulating double-loop and transformative learning from discontinuous events, both for managers and for entrepreneurs.

This article has focused on a special 'breed' of managers, namely individuals who are embroiled in the ownership and management of small enterprises. It has been suggested that the study of entrepreneurs represents a rich, unique context in which to explore the phenomenon of management learning. This is due to the complexity of the relationship between entrepreneurs and their businesses and the extreme levels of personal investment, resources and exposure that entrepreneurs are faced with in starting and managing a small venture. This multifaceted and emotional relationship has significant implications for making sense of the learning outcomes associated with discontinuous events. For, as Brown (2000) remarks, 'learning and emotion work together in a tandem motion' (p. 290). An interesting avenue for further research involves exploring the affective dimension of these events and the relationship between the emotional 'intensity' of the event and the associated depth of personal reflection and learning. It would be particularly beneficial to explore this issue in relation to what is, arguably, the most extreme organizational discontinuous event faced by entrepreneurs, business failure.

Finally, although this article reports on learning experienced by individuals working within the specific context of small business ownership, it must be emphasized that the learning outcomes described here are not confined necessarily to entrepreneurs. This recognition reflects the idea that people, in general, tend to think back and make sense of memorable events that happen in their lives and that such a reflective process can give rise to fundamental and challenging forms of learning. In this regard, discontinuous events continue to be an interesting and necessary phenomenon in understanding how individuals learn.

Notes

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1. Juxtaposing Mezirow's (1990, 1991) concept of instrumental learning with the work of Fiol and Lyles (1985), such learning could be described as 'lower-level' because it involves the detection and correction of error and the modification of *existing* procedures, strategies, actions and assumptions.

2. The work of Bateson (1973) suggests *three* rather than two levels of learning. However, on empirical analysis within the field of management learning, Burgoyne and Hodgson (1983) equate Bateson's 'Level 1' and 'Level 2' learning with Argyris and Schön's (1978) 'single-loop' learning. In contrast, 'Level 3' learning is equated with 'double-loop' learning because it has the capacity to affect 'deeper underlying values, concepts and orientations' (Burgoyne and Hodgson, 1983: 398). In so doing, Burgoyne and Hodgson (1983) reinforce the dualistic nature of learning levels.
3. For a more in-depth discussion of the research design and some of the broader findings from the study refer to Cope and Watts (2000).
4. All the participants discussed in this article have been given pseudonyms.
5. In terms of definition, both case examples illustrate that a 'discontinuous event' is not always as discrete or isolated as the term may suggest. It is apparent that such non-routine events can take place over a period of time and it often proves very difficult, both for the researcher and the researched, to define precisely the chronological and perceptual boundaries of these events (Cope and Watts, 2000). It is also clear from both case examples that discontinuous events do not occur 'out of the blue'; rather they are often inextricably linked to a complex set of circumstances and actions.
6. Argyris and Schön (1978) define 'theory-in-use' (as opposed to 'espoused theory') as a complex system of norms, rules and assumptions that are constructed through the observation of actual behaviour; norms that are implicit and ingrained within the organization. As such, this 'theory' of how individual members are meant to act remains largely tacit and unspoken.
7. This statement by Luke raises some extremely interesting issues regarding the potential blurring of boundaries between individual and organizational learning in smaller firms. However, it is not the intention of this article to engage in a debate regarding the relationship between individual versus organizational learning, although it is clear that this issue continues to attract significant attention (see, for example, Prange, 1999). Rather, it is the *individual* learning outcomes experienced by entrepreneurs as a result of facing discontinuous events that remains the central theme of this article.

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